O Mirror, Where Art Thou?

Position retrieval using nested sampling and global minimization

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Motivation

Can we tell where a telescope's mirrors are from its images alone?

A misaligned mirror distorts the point spread function (PSF) differently across the field. Phase retrieval recovers the wavefront, not the mirror positions.

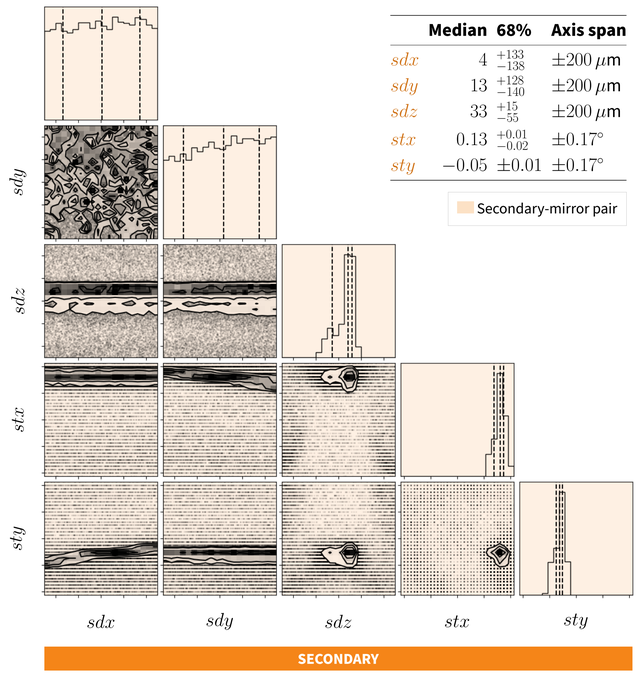
What we did: infer rigid-body mirror positions *directly* from multi-field PSFs.

- Ray-traced *Hubble* (5 secondary-mirror DoF) and a compact TMA (10 primary + secondary DoF) in Optiland [1].
- Compared multi-field PSF coefficients via a Gaussian likelihood.
- Mapped the posterior with nested sampling, then refined it with differential evolution.
- Checked each solution by the residual wavefront error.

Validating on *Hubble*

Ritchey–Chrétien model (2.4 m pupil, $\lambda = 0.55\,\mu$m) with five secondary-mirror degrees of freedom: decenter (sdx, sdy), despace sdz and tilts (stx, sty). Noise-free PSFs at 5 fields, 15 circular-Bessel modes, priors of $\pm 200\,\mu$m and $\pm 0.167°$.

- Nautilus (42 800 likelihood calls, $\ln \mathcal{Z} \approx +107.5$) tightly constrains both tilts and, more loosely, sdz.
- sdx and sdy stay nearly flat across the $\pm 200\,\mu$m prior.
- DE starts from the median of the final 100 posterior samples, $(131.13,\ 56.64,\ 40.87,\ 0.12,\ -0.03)$, which alone cuts the on-axis OPD RMS from 2.26 to 0.12 waves.

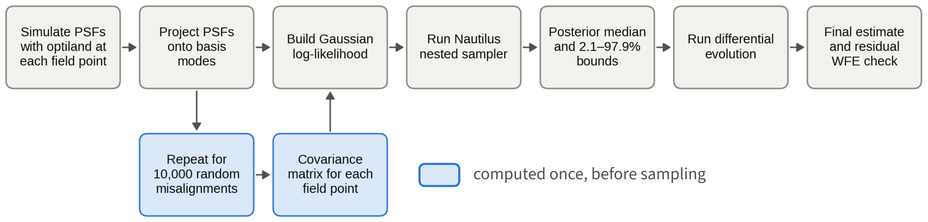


	Median	68%	Axis span
sdx	4	$^{+133}_{-138}$	$\pm 200\,\mu$m
sdy	13	$^{+128}_{-140}$	$\pm 200\,\mu$m
sdz	33	$^{+15}_{-55}$	$\pm 200\,\mu$m
stx	0.13	$^{+0.01}_{-0.02}$	$\pm 0.17°$
sty	−0.05	± 0.01	$\pm 0.17°$

Nautilus posterior for the five secondary-mirror parameters; the table gives full-posterior medians and 68% intervals.

Even so, DE using only the on-axis field recovers every translation to within $0.01\,\mu$m of the truth (centre panel).

Inference pipeline



Optiland [1] PSF coefficients enter a Gaussian likelihood:

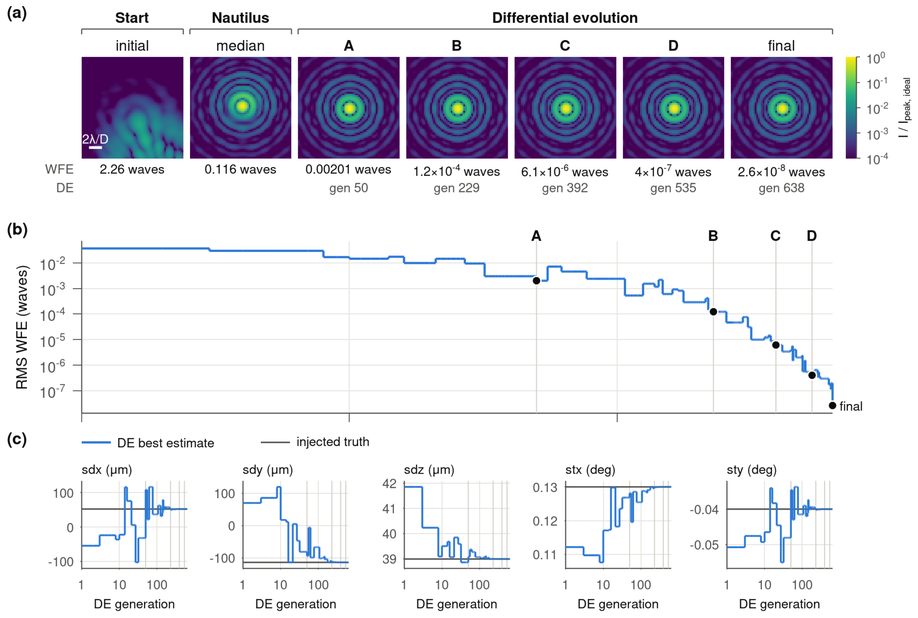
$$\ln \mathcal{L}(\boldsymbol{\theta}) = \sum_{f \in \mathcal{F}} -\tfrac{1}{2}\,\Delta\mathbf{c}_f^{\top}\,\boldsymbol{\Sigma}_f^{-1}\,\Delta\mathbf{c}_f + \text{const},$$

$$\Delta\mathbf{c}_f = \mathbf{c}_f(\boldsymbol{\theta}) - \mathbf{c}_{f,\text{obs}}$$

$\boldsymbol{\Sigma}_f$ from 10^4 random misalignments.

- **Nested sampling** (Nautilus [2,3]) maps the posterior.
- **Differential evolution** [4,5] refines the Nautilus starting point.
- **Validation:** on-axis OPD RMS of the residual telescope.

Algorithm recovers *Hubble*'s optical alignment to numerical precision



(a) On-axis PSF, uncorrected (2.26 waves) to final DE solution (2.6×10^{-8} waves). **(b)** RMS wavefront error per DE generation. **(c)** Parameters converging to the injected truth.

Translations recovered to $< 0.01\,\mu$m:

	Truth	Recovered (DE)
sdx (μm)	52.22	52.22
sdy (μm)	−113.86	−113.86
sdz (μm)	38.99	38.99
stx (deg)	0.13	0.13
sty (deg)	−0.04	−0.04

References

[1] Harrison 2026, Optiland v0.6.0

[2] Lange 2023, MNRAS 525, 3181

[3] Skilling 2006, Bayesian Analysis 1, 833

[4] Storn & Price 1997, J. Global Optim. 11, 341

[5] Virtanen et al. 2020, Nature Methods 17, 261

[6] Kim, Choi & Douglas 2023, Proc. SPIE 12677, 126770E

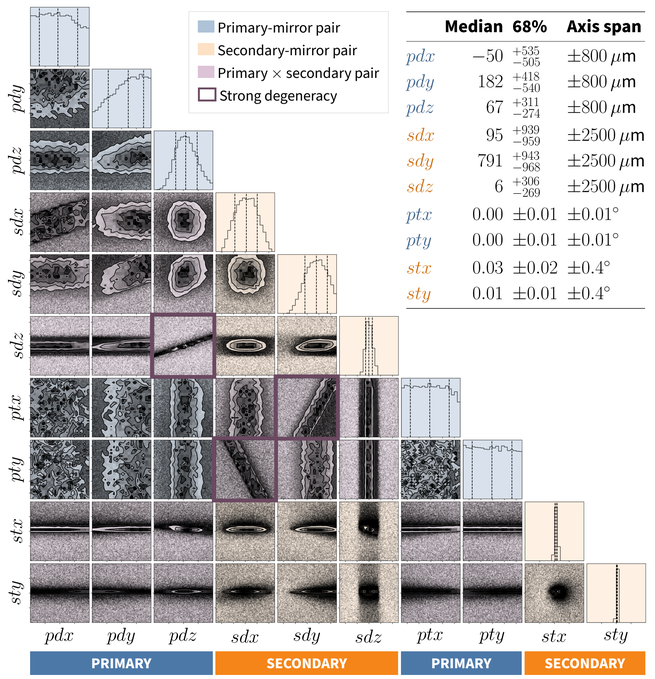
We thank T. Eifler, N. Franz, K. Derby and S. Blomquist.

Validating on a TMA

Compact TMA [6] (6.47 m pupil, $\lambda = 0.65\,\mu$m) with ten degrees of freedom: decenter, despace and tilt of both the primary and the secondary. PSFs at 10 field points carry Gaussian noise ($\sigma = 0.01$ per pixel) and are projected onto 11 Zernike modes.

- Nautilus: 93 200 likelihood calls, $\ln \mathcal{Z} \approx +484.7$.
- DE recovers all translations to $\lesssim 17\,\mu$**m** and reduces on-axis OPD RMS from 8.24 to 7×10^{-3} waves.

	Primary		Secondary	
	Truth	Recovered	Truth	Recovered
dx (μm)	−152.75	−159.43	−42.31	−38.01
dy (μm)	316.91	309.97	923.33	939.85
dz (μm)	26.63	24.77	−32.75	−34.56
tx (deg)	2.98×10^{-5}	2.22×10^{-4}	0.0269	0.0272
ty (deg)	1.61×10^{-5}	6.80×10^{-5}	0.0098	0.0098



	Median	68%	Axis span
pdx	−50	$^{+535}_{-505}$	$\pm 800\,\mu$m
pdy	182	$^{+418}_{-540}$	$\pm 800\,\mu$m
pdz	67	$^{+311}_{-274}$	$\pm 800\,\mu$m
sdx	95	$^{+939}_{-959}$	$\pm 2500\,\mu$m
sdy	791	$^{+943}_{-968}$	$\pm 2500\,\mu$m
sdz	6	$^{+306}_{-269}$	$\pm 2500\,\mu$m
ptx	0.00	± 0.01	$\pm 0.01°$
pty	0.00	± 0.01	$\pm 0.01°$
stx	0.03	± 0.02	$\pm 0.4°$
sty	0.01	± 0.01	$\pm 0.4°$

Nautilus posterior for the ten TMA parameters, shaded by mirror. Outlined panels are the strongest primary–secondary degeneracies.

- pdx and both primary tilts are nearly flat across their priors; only the secondary tilts are tightly constrained.
- Degeneracies: pdz–sdz trade off in focus; ptx–sdy and pty–sdx trade off in coma, because a tilt of one mirror and a decenter of the other both add nearly field-constant coma.

Conclusions and next steps

- Mirror alignment is recovered directly from multi-field PSF coefficients.
- ***Hubble*:** OPD RMS 2.26 → 3×10^{-8} waves.
- **TMA:** OPD RMS 8.24 → 7×10^{-3} waves.
- Degeneracies motivate multi-field sensing.
- Next: chromatic and realistic noise models, and laboratory and on-sky PSF data for future large space telescopes.